Multivariate Behavioral Research

ISSN: 0027-3171 (Print) 1532-7906 (Online) Journal homepage: www.tandfonline.com/journals/hmbr20

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To cite this article: Edward E. Rigdon, Jan-Michael Becker & Marko Sarstedt (2019) Factor Indeterminacy as Metrological Uncertainty: Implications for Advancing Psychological Measurement, Multivariate Behavioral Research, 54:3, 429-443, DOI: 10.1080/00273171.2018.1535420

To link to this article: https://doi.org/10.1080/00273171.2018.1535420

Published online: 04 Mar 2019.

MULTIVARIATE BEHAVIORAL RESEARCH
2019, VOL. 54, NO. 3, 429–443
https://doi.org/10.1080/00273171.2018.1535420

Factor Indeterminacy as Metrological Uncertainty: Implications for Advancing Psychological Measurement

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ABSTRACT

Researchers have long been aware of the mathematics of factor indeterminacy. Yet, while occasionally discussed, the phenomenon is mostly ignored. In metrology, the measurement discipline of the physical sciences, uncertainty – distinct from both random error (but encompassing it) and systematic error – is a crucial characteristic of any measurement. This research argues that factor indeterminacy is uncertainty. Factor indeterminacy fundamentally threatens the validity of psychometric measurement, because it blurs the linkage between a common factor and the conceptual variable that the factor represents. Acknowledging and quantifying factor indeterminacy is important for progress in reducing this component of uncertainty in measurement, and thus improving psychological measurement over time. Based on our elaborations, we offer a range of recommendations toward achieving this goal.

KEYWORDS
Structural equation modeling; factor indeterminacy; validity; measurement models

"Messen ist Wissen." ("Measurement is Knowledge.")

Georg Simon Ohm (1789–1854), quoted in Goldsmith (2010)

Introduction

As one aspect of the multivariate method known as factor analysis, factor indeterminacy has received occasional attention in the literature, across a span ranging from Wilson's (1928) introduction of the problem, through Guttman's (1955) seminal treatment, through a burst of attention in the 1970s (Mulaik & MacDonald, 1978; Schönemann & Steiger, 1978; Schönemann & Wang, 1972; among others), a 1996 special section in *Multivariate Behavioral Research* (focal article by Maraun, 1996, with multiple rounds of responses by multiple authors), a readable treatment by Grice (2001), and others. What Schönemann and Steiger (1978) and Steiger (1979b) observed 40 years ago summarizing the factor analysis literature still holds today: the factor analysis literature commonly concludes that factor indeterminacy does not matter, is essentially irrelevant, and is not even worth quantifying or reporting. This dismissal of the factor indeterminacy phenomenon may be best summed up by Hoyle's (2014) *Handbook of Structural Equation Modeling* – across the 700+ pages of this volume, the only reference to factor indeterminacy is where Hoyle (2014, p. 4) judges it to be "more an inconvenience than a limitation."

Most likely, this judgment follows from what factor indeterminacy does *not* do. It does not directly affect parameter estimates within a particular model, or the model's standard errors, or the model's fit indices. And yet, this paper argues that managing factor indeterminacy, and working to minimize it over time, is an important key to progress in psychological measurement. This paper supports this argument using concepts drawn from metrology – measurement science in the physical and life sciences and engineering – and philosophy of science. In particular, the metrological concept of uncertainty and the philosophical stance known as scientific realism point to the management and minimization of factor indeterminacy as crucial to progress in psychometrics.

The remainder of the paper begins with a brief review of the mathematics of factor indeterminacy. Then it introduces metrology and specifically the central concept of uncertainty as understood in that field,

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with the aim of showing that factor indeterminacy is indeed a form of uncertainty. Then the paper briefly introduces scientific realism, a philosophy of science, which asserts that the unobserved conceptual variables in theoretical models are defeasibly real and are subject to the scientific method. The paper then argues that the uncertainty due to factor indeterminacy applies to the relation between a common factor and the conceptual variable that the factor represents, making factor indeterminacy not just a nuisance, but a threat to validity in psychological measurement. The paper then concludes with recommendations for moving psychological measurement forward.

Within this paper, the term "conceptual variable" is taken to mean an attribute, like "job satisfaction" or "organizational commitment," of an individual, group, or system. Conceptual variables may in principle be observable but they are most often unobservable. Conceptual variables populate theoretical models in the behavioral sciences, and are believed to have causal power over other observed and unobserved variables, which is why researchers study them. A common factor, by contrast, is a variable in a statistical model, which is not directly observed. Other literature has used terms like "construct" or "latent variable" to describe both the conceptual variables in theoretical models and the common factors in statistical models (Maraun & Gabriel, 2013; Markus & Borsboom, 2013; Michell, 2013). This paper will always distinguish between conceptual variables and common factors.

Factor indeterminacy

A common factor model specifies that a set of P observed variables $\boldsymbol{Y}_P$ (y_p, $p = 1, \ldots, P$), are dependent on K common factors $\boldsymbol{F}_K$ (f_k, $k = 1, \ldots, K$), one for each conceptual variable T_k, if the model is linear, and P residuals or unique factors $\boldsymbol{U}_P$ (u_p, $p = 1, \ldots, P$, one for each observed variable y_p), according to P equations of the form:

$$y_p = \lambda_{pk} f_k + u_p, \tag{1}$$

which can be gathered into one matrix equation of the form:

$$Y_{P\times1} = \Lambda_{P\times K} F_{K\times1} + U_{P\times1}, \tag{2}$$

The loading of each unique factor u_p on its associated observed variable y_p is implicitly 1. Making this explicit, the model might also be depicted with an identity matrix of loadings for $\boldsymbol{U}$:

$$Y_{P\times1} = \Lambda_{P\times K} F_{K\times1} + \mathrm{I}_{P\times P} U_{P\times1}. \tag{3}$$

Equation (4) expresses this model by combining $\boldsymbol{F}$ and $\boldsymbol{U}$ into one partitioned vector:

$$Y_{P\times1} = \begin{bmatrix} \Lambda_{P\times K} & \mathrm{I}_{P\times P} \end{bmatrix} \begin{bmatrix} F_{K\times1} \\ U_{P\times1} \end{bmatrix} \tag{4}$$

with expected values $\mathrm{E}(y_p) = \mathrm{E}(f_k) = \mathrm{E}(u_p) = 0$, for all p and all k. With the standard assumption that the common factors $\boldsymbol{F}$ are orthogonal with the unique factors $\boldsymbol{U}$ (e.g., Bollen, 1989, p. 233), the covariance structure for the observed variables becomes:

$$\Sigma_{P\times P} = E(YY') = \begin{bmatrix} \Lambda & \mathrm{I} \end{bmatrix}_{P\times(P+K)} \begin{bmatrix} \Phi_{K\times K} & 0 \\ 0 & \Psi_{P\times P} \end{bmatrix} \begin{bmatrix} \Lambda' \\ \mathrm{I} \end{bmatrix}_{(K+P)\times P},$$

$$\Sigma_{P\times P} = E(YY')$$

$$= \begin{bmatrix} \Lambda & \mathrm{I} \end{bmatrix}_{P\times(P+K)} \mathrm{W}_{(P+K)\times(P+K)} \begin{bmatrix} \Lambda' \\ \mathrm{I} \end{bmatrix}_{(K+P)\times P}. \tag{5}$$

Here, $\boldsymbol{\Sigma}$ is the covariance matrix of the observed variables, $\boldsymbol{\Phi}$ is the covariance matrix of the common factors, and $\boldsymbol{\Psi}$ is the covariance matrix of the unique factors. The partitioned matrix $\mathbf{W}$ combines $\boldsymbol{\Phi}$ and $\boldsymbol{\Psi}$, with 0 cross-covariances reflecting the orthogonality constraint. This development assumes that the factor model is strictly correct and that model parameters are population values.

In simplest terms, "[f]actor indeterminacy is the inability to determine uniquely the common and unique factor variables of the common factor model from the uniquely defined 'observed variables' because the number of observed variables is smaller than the number of common and unique factors" (Mulaik & McDonald, 1978, p. 177). The phenomenon can also be understood in terms of the ranks of $\boldsymbol{\Sigma}$ and $\mathbf{W}$, the covariance matrices on each side of Equation (5). $\boldsymbol{\Sigma}$, the covariance matrix of the P observed variables, has rank at most P, meaning that there are at most P distinct dimensions of information in the data (Mulaik, 2010). Yet common factor models normally exceed this limit: on the right hand side, $\mathbf{W}$ will typically have rank $P + K$. The common factor model turns the maximum P dimensions of information in the data into $P + K$ dimensions of information (Schönemann & Steiger, 1978, pp. 287–288, derived the same conclusion using vector spaces). This inflation of information is accomplished by making the common factors and unique factors indeterminate.

Guttman (1955) demonstrated that an equation of the form of Equation (5) can be solved for the common and unique factors. This leads to standard solutions (see Guttman, 1955, p. 74, Equations 46, 47):

$$\begin{aligned} F_{K\times1} &= \Phi\Lambda'\Sigma^{-1}Y + \mathrm{B}_{K\times K} Z_{K\times1}, \\ U_{P\times1} &= \Psi\Sigma^{-1}Y - \Lambda\mathrm{B}_{K\times K} Z_{K\times1}. \end{aligned} \tag{6}$$

These $\boldsymbol{F}$ and $\boldsymbol{U}$ are *not* mere "estimates" or approximations of the common and unique factors. They are precisely the same common and unique factors from the original factor model.

The new elements in these equations are **B** and ***Z***. The $K \times K$ matrix **B** is a "Gram factor," a matrix such that **BB**' = **M** for some matrix **M**. Here, **M** is (see Guttman, 1955, p. 74, Equation 50; Schönemann & Steiger, 1976, p. 182, Equation 2.18):

$$\mathrm{BB}' = \mathrm{M} = \Phi - \Phi\Lambda'\Sigma^{-1}\Lambda\Phi. \tag{7}$$

The matrix product $\mathbf{\Phi\Lambda'\Sigma^{-1}\Lambda\Phi}$ is equal to σ^2 ($\boldsymbol{F}$ | $\boldsymbol{Y}$), the conditional variance/covariance matrix of the common factors ***F*** given the observed variables ***Y*** (Mulaik, 2010, p. 380). In other words, this is the part of the covariance matrix of the common factors that can be explained or accounted for by the observed variables. With standardized population data, the diagonal elements of this matrix product will equal $\rho^2_{f.Y}$, the squared multiple correlation of each common factor as predicted by all of the model's observed variables (Mulaik, 2010, p. 380). Since $\mathbf{\Phi}$ is the covariance matrix of the common factors, $\mathbf{BB}' = \mathbf{M}$ represents the part of the covariance matrix of the common factors which is *not* explained by the observed variables. Thus, the elements of **B** are weighting coefficients, much like reliabilities.

Z is a set of K independent variables z_1, z_2 … z_K (Guttman, 1955, p. 72), one for each common factor. Besides being orthogonal among themselves, the z_k must be uncorrelated with all other variables *included* in the factor model (except for the factor of which each z_k itself forms a part). Significantly, there are no explicit constraints on covariances between any z_k and any variable, which is *not* included in the factor model (Steiger, 1979a). As a reviewer points out, correlations between the variables in ***Z*** and variables outside the model *are* implicitly constrained to the extent that the outside variables correlate with variables within the model. Such correlations will be uncertain, but will be constrained by the correlations between the outside variable and the variables within the factor model. Subject to these constraints, there are infinitely many possible values for ***Z*** for any model with indeterminate factors. Correctly constructed, all of these different versions of ***Z*** will produce common and unique factors that are completely consistent with the factor model and all of its constraints, even though the different instances of a given common or unique factor need not be highly correlated among themselves (Guttman, 1955).

The mathematics of factor indeterminacy, a phenomenon first noted by the mathematician Wilson (1928), are not in dispute. The question is, what does factor indeterminacy mean for researchers? As previously acknowledged, factor indeterminacy does not induce bias in parameter estimates, standard errors, or fit indices. Given that, researchers may well feel that the phenomenon can safely be ignored. However, past examinations of this issue have not been informed by developments in measurement science outside psychometrics, nor by the philosophy of scientific realism. The paper next turns to these topics and their implications.

Metrology and its implications for factor indeterminacy

Uncertainty of measurement

Metrology is "the science of measures or measurements traceable to measurement standards" (Slaev, Chunovkina, & Mironovsky, 2013, p. vii). Across the biological and physical sciences and engineering, metrology fills the same role that psychometrics fills for the behavioral sciences. Metrology defines measurement in the classical sense, as a matter of ratios – the length of an object in meters is the ratio of the object's length to the length of one meter (Michell, 1999). Metrology also defines measurement as involving instruments for capturing data. Measurement of attributes which cannot be directly captured by instruments, like the physical sensations of human taste, touch or smell, lie outside formal metrology and within the emerging field of "soft metrology" (Pointer, 2003). Metrology shies away from measuring purely mental phenomena, which cannot be directly captured by objective instrument, though soft metrology aims to measure physical attributes, like heart rate, which may be linked to mental phenomena.

A fundamental concept in metrology is the uncertainty of measurement (JCGM WG1, 2008, section 2.1). Uncertainty is formally defined as:

> Parameter, associated with the result of a measurement, that characterizes the dispersion of the values that could reasonably be attributed to the measurand (JCGM WG1, 2008, section 2.2.3).

That marks an important distinction from statistics like standard errors or confidence intervals, which describe the dispersion of values of the statistic. For example, a 95% confidence interval is an interval such that 95% of all such calculated intervals will include those values. It is not – despite many erroneous interpretations – an interval 95% likely to contain the true value being estimated (e.g., Mayo, 1981, p. 272). Uncertainty by definition covers the plausible range of values for the measurand.

A statement about uncertainty is an essential component of any measurement:

> In general, the result of a measurement … is only an approximation or estimate of the value of the measurand and thus is complete only when accompanied by a statement of the uncertainty (…) of that estimate (JCGM WG1, 2008, section 3.1.2).

Uncertainty can include contributions from multiple sources, some based on statistics and some based on judgment (Bucher, 2012).

In metrology, uncertainty and random variance are *not* the same thing:

> In this Guide, great care is taken to distinguish between the terms 'error' and 'uncertainty.' They are not synonyms, but represent completely different concepts; they should not be confused with one another or misused (JCGM WG1, 2008, section 3.2.2, NOTE 2).

Uncertainty encompasses random error – random error from various sources may contribute to uncertainty – as one component, but uncertainty does not behave like random error. Random errors offset each other: increasing sample size reduces random sampling error, even when drawing from an infinite population. By contrast, uncertainties accumulate (e.g., Bucher, 2012, Ch. 29). Suppose a metrologist is measuring the output of a certain process using a scale. Perhaps the reading from the scale is affected by some uncertainty due to human variation but the scale is also affected by ambient temperature, which itself is assessed by a gauge that carries its own uncertainty. In some cases, uncertainty is a result of the limited precision of measurement for some component of a device or operation. If the gauge described here is only denominated in whole degrees, for example, then there will be uncertainty equal to about one degree in the measurement of ambient temperature, and this uncertainty would become a part of the uncertainty associated with the measurement of the output of the process.

Metrologists combine these distinct sources of uncertainty through "addition by quadrature" or root sum of squares – squaring the individual uncertainties, summing the squares, then taking the square root (Bell, 1999, p. 14; Bucher, 2012, Ch. 29). An important task for the metrologist is to identify all of the material contributors to the uncertainty associated with a measurement, so as to be able to state the total uncertainty. What is material depends on the need of the task at hand: "So the key challenge for metrologists here is not to get the right answer, but to decide on the uncertainty they can tolerate" (Bell, 1999, p. 19).

Uncertainty afflicts measurement at even the most sophisticated levels as it blurs the values of measurements, and then blurs any parameter estimates based on those measures – researchers simply lose a degree of precision (e.g., Tal, 2011, p. 1090). Even those metrologists who are charged with precisely measuring (unobservable) *time* using cesium fountain atomic clocks must address both error *and* uncertainty of their measurements (Tal, 2011): different cesium fountain clocks will not agree precisely, nor can precise agreement be achieved through simple adjustments.

To gradually improve the quality of any measurement, metrologists build "uncertainty budgets" for their measurements, cataloging all the substantial sources of uncertainty (e.g., Bucher, 2012; Tal, 2011). For example, Tal (2011, p. 1094; see also Rosenband et al., 2008) establishes an uncertainty budget comparison of two atomic clocks. The uncertainty budget lists 11 items, including "micromotion," "black body radiation," "background gas collision," and "gravitational redshift." In the measurement of time, efforts to reduce uncertainty have resulted in substantial quality gains. In the mid-1600s, the best available clock, with a pendulum design, lost no more than 10 seconds per day (Higgins, Miner, Smith, & Sullivan, 2004). By 1889 – not long before Spearman (1904) launched modern psychometrics – the best clock was accurate to a tenth of a second per day (Higgins et al., 2004). By 2002, a cesium fountain clock at the National Institute of Standards and Technology was rated accurate to within 30 billionths of a second per year (Higgins et al., 2004) On today's cutting edge, "optical" clocks (which monitor visible spectrum light rather than the microwave radiation captured by cesium fountain clocks) promise such accuracy that two of them would not disagree by as much as 1 second over the entire life of the universe (Gibney, 2015). The latest devices give physicists a new tool to test old implications of relativity theory, including "time dilation," the hypothesis that large gravitational bodies, like planets, change the speed at which time passes. Pairs of these clocks make time dilation detectable even when the difference in height (that is, the difference in distance from the center of the earth) of the two clocks is less than one meter (Chou, Hume, Rosenband, & Wineland, 2010).

Uncertainty and factor indeterminacy

Different from metrology, the psychometric measurement framework which dominates the social sciences, and which is closely tied to factor analysis (McDonald, 1999; Schönemann & Steiger, 1978), does not recognize uncertainty as a distinct measurement phenomenon. Psychometrics addresses only error – random error and systematic error. So what does factor indeterminacy represent from a psychometric perspective?

If factor indeterminacy was a matter of random variance, then relations of indeterminate common factors with outside variables – variables not a part of the statistical model containing those common factors – would simply be attenuated in familiar ways (e.g., Rigdon, 1994). Factor indeterminacy does not affect parameter estimates *within* the model at all. Returning to the mathematical discussion of factor indeterminacy, recall that there is one variable in $\boldsymbol{Z}$ for each indeterminate common factor. While it is possible to construct $\boldsymbol{Z}$ entirely out of random variance, the variance can also be systematic, as long as each variable in $\boldsymbol{Z}$ meets the specified requirements (Guttman, 1955; Steiger, 1979a). While the variables in $\boldsymbol{Z}$ are constrained to orthogonality *within* the common factor model (they cannot correlate among themselves nor with any common factor of which they are not a part), they can be positively or negatively correlated – even perfectly correlated – with any variable *not* included in the factor model, though these correlations again, will be effectively constrained to the extent that the outside variable is itself correlated with variables contained within the factor model. The correlation between an indeterminate common factor and an outside variable is a function of both the degree of indeterminacy in the common factor and the multiple correlation between the observed variables in the model (which form the determinate part of the factor) and the outside variable (Steiger, 1979a).

Let such an outside variable be labeled ω. Then the correlation between a particular common factor f and ω will fall in some range whose width, κ, is equal to (Steiger, 1979a, p. 95):

$$\kappa = 2\sqrt{1-\rho^2_{f.Y}}\sqrt{1-\rho^2_{\omega.Y}}, \tag{8}$$

where, as before, $\rho^2_{f.Y}$ is the squared multiple correlation of the factor f predicted by the model's observed variables and $\rho^2_{\omega.Y}$ is the squared multiple correlation of the outside variable predicted by all of the observed variables in the model. So the indeterminacy-imposed width of the range of the correlation between common factor and outside variable is a function of two squared multiple correlations.

The second coefficient, $\rho^2_{\omega.Y}$, is generally unknowable, but it is possible to develop some intuition about its likely value. First, adding more observed variables to $\boldsymbol{Y}$ cannot lower $\rho^2_{\omega.Y}$ and should tend to raise it. With sample data, the multiple correlation in a regression will always tend to increase due to sampling error (e.g., Cohen, Cohen, West, & Aiken, 2003), but that should not necessarily be expected at the level of the population, as a reviewer helpfully noted. Still, researchers could expect a higher value when there are many observed variable indicators in the common factor model. Example calculations stretching as far back as Guttman (1955) illustrate this effect. While it is certainly possible (as a reviewer pointed out) that $\rho^2_{\omega.Y} > \rho^2_{f.Y}$, it would be a challenging position to justify in any given instance, lacking any specific information. The observed variables are assumed to be contaminated with random error – that is part of the rationale for using a factor model – and that will attenuate all of the observed variables' correlations – those with ω as much as those with f. Moreover, given that the observed variables were intended as indicators in a factor model, they will have been designed to be somewhat redundant, whereas maximizing prediction of some criterion would have been best served by using nonredundant predictors. It is certainly true that, in a model with many common factors, the entire set of observed variables is not collectively unidimensional, so a researcher might expect better prediction of an arbitrary criterion when a model includes many factors. Then again, the researchers might have been entirely wrong in their research design, and the observed variables might have nothing to do with a given conceptual variable except for some spurious correlation. Still, in a given instance, with the actual value unknown, a common sense *best-case scenario* would set the two squared multiple correlations in Equation (8) equal to each other, so the formula for a simplified $\kappa_{=}$ would be:

$$\kappa_{=} = 2\left(1 - \rho^2_{f.Y}\right). \tag{9}$$

Guttman (1955) derived an indeterminacy index, ρ_{min}, equal to the lowest possible (being in some cases negative) correlation between two different sets of factor scores (i.e., numbers assigning a specific value to a common factor for each respondent) for the same common factor, model, and data:

$$\rho_{min} = 2\rho^2_{f.Y} - 1. \tag{10}$$

The simplified $\kappa_{=}$, based on Steiger (1979a), proves to be equal to 1 – ρ_{min}.

The first coefficient in Equation (8), $\rho^2_{f.Y}$, can be estimated from standardized parameter estimates and the correlation matrix of the observed variables, using the formula:

$$R^2_{F.Y} = diag\left(\Phi_s \Lambda'_s R_Y^{-1} \Lambda_s \Phi_s\right), \tag{11}$$

where, the subscript s indicates standardization, and $\mathbf{R_Y}$ is the correlation matrix of the observed variables.[1] For example, imagine a common factor model with eight observed variable indicators y_p $(p = 1, \ldots, 8)$ with $y_1 - y_4$ loading on one common factor and $y_5 - y_8$ loading on another common factor. Let all standardized loadings be equal to 0.80 and all unique factor variances be equal to 0.36, and let the correlation between the common factors equal 0.50. The population correlation matrix implied by this factor model has lower triangular form:

$$R_Y = \begin{matrix} 1 \\ .64 & 1 \\ .64 & .64 & 1 \\ .64 & .64 & .64 & 1 \\ .32 & .32 & .32 & .32 & 1 \\ .32 & .32 & .32 & .32 & .64 & 1 \\ .32 & .32 & .32 & .32 & .64 & .64 & 1 \\ .32 & .32 & .32 & .32 & .64 & .64 & .64 & 1 \end{matrix}$$

Then:

$$R^2_{F.Y} = diag\left(\Phi_s \Lambda'_s R_Y^{-1} \Lambda_s \Phi_s\right) \approx \begin{matrix} .88 & \\ & .88 \end{matrix}$$

So $\rho^2_{f.Y} \approx .88$, and the best-case estimate for the width of the range is $2(1 - \rho^2_{f.Y}) = 2(1 - .88) = .24$. Because of factor indeterminacy, the correlation between common factor and outside variable lies in a range whose width is 0.24. This is not like a confidence interval, where a range of possible outcomes is distributed around a central tendency. All correlation values within a range of this width would be equally plausible – the distribution of plausible correlation values is uniform across this range. This uncertainty is a consequence of the indeterminacy, which is inherent in the common factor model. The greater the residual variance in the indicators, the lower the loadings, and the fewer the indicators, the greater the factor indeterminacy and the resulting uncertainty. Increasing the sample size does not reduce this uncertainty – it remains even at population level. Factor indeterminacy *is* uncertainty.

Tables 1–3 demonstrate the impact of varying numbers of indicators and factors, and varying standardized loadings and factor correlations, on both $\kappa_=$ and ρ_{min}. Recall that $\kappa_=$ is the range of uncertainty around a correlation – the correlation between a common factor and any variable outside the model. For example, a model with one factor and two indicators with standardized loadings of 0.50 produces $\kappa_= = 1.200$, which means that the range of uncertainty could cover 60% of the total possible (+1 – -1 = 2) range of a correlation (Table 1). On the other hand, with four common factors in the model, eight indicators per factor, standardized loadings of 0.90, and factor correlations of 0.70 (Table 3), $\kappa_=$ is 0.055, indicating a small range of uncertainty.

Table 1. Uncertainty $\kappa_=$ and Guttman's ρ_{min}: One Common Factor, Equal Loadings.

	Number of indicators				
	2	3	4	6	8
Loading					
0.50	1.200	1.000	0.857	0.667	0.544
	(−0.200)	(0.000)	(0.143)	(0.333)	(0.456)
0.60	0.951	0.744	0.615	0.457	0.364
	(0.059)	(0.256)	(0.385)	(0.543)	(0.636)
0.70	0.685	0.515	0.413	0.296	0.230
	(0.315)	(0.485)	(0.587)	(0.704)	(0.770)
0.80	0.439	0.316	0.247	0.171	0.131
	(0.561)	(0.684)	(0.753)	(0.829)	(0.869)
0.90	0.210	0.145	0.111	0.075	0.057
	(0.790)	(0.855)	(0.889)	(0.925)	(0.943)

Tabled values are values of $\kappa_=$ with Guttman's ρ_{min} in parentheses.

Table 2. Uncertainty $\kappa_=$ and Guttman's ρ_{min}: Four Common Factors, Equal Loadings, factor correlations all 0.3.

	Number of indicators per factor				
	2	3	4	6	8
Loading					
0.50	1.135	0.945	0.812	0.635	0.523
	(−0.135)	(0.055)	(0.188)	(0.365)	(0.477)
0.60	0.890	0.707	0.582	0.440	0.352
	(0.110)	(0.293)	(0.412)	(0.560)	(0.648)
0.70	0.652	0.495	0.399	0.288	0.225
	(0.348)	(0.505)	(0.601)	(0.712)	(0.775)
0.80	0.423	0.307	0.241	0.169	0.130
	(0.577)	(0.693)	(0.759)	(0.831)	(0.870)
0.90	0.206	0.143	0.110	0.075	0.057
	(0.794)	(0.857)	(0.890)	(0.925)	(0.943)

Tabled values are values of $\kappa_=$ with Guttman's ρ_{min} in parentheses.

Table 3. Uncertainty $\kappa_=$ and Guttman's ρ_{min}: Four Common Factors, Equal Loadings, factor correlations all 0.7.

	Number of indicators per factor				
	2	3	4	6	8
Loading					
0.50	0.880	0.724	0.623	0.496	0.417
	(0.120)	(0.276)	(0.377)	(0.504)	(0.583)
0.60	0.682	0.547	0.463	0.359	0.295
	(0.318)	(0.453)	(0.537)	(0.641)	(0.705)
0.70	0.508	0.397	0.329	0.247	0.199
	(0.492)	(0.603)	(0.671)	(0.753)	(0.801)
0.80	0.347	0.261	0.211	0.153	0.120
	(0.653)	(0.739)	(0.789)	(0.847)	(0.880)
0.90	0.183	0.131	0.102	0.071	0.055
	(0.817)	(0.869)	(0.898)	(0.929)	(0.945)

Tabled values are values of $\kappa_=$ with Guttman's ρ_{min} in parentheses.

[1]Here, parameter estimates replace population values as necessary.

Scientific realism and the nature of common factors

An important further implication of this identification of factor indeterminacy as metrological uncertainty depends upon one's perspective on philosophy of science. Different approaches to philosophy of science are complex and subtle – the argument here is a matter of broad strokes.

In the first half of the 1900s, the social sciences were dominated by the philosophy of logical empiricism (Fine, 1998; Keat, 1998). Of course, like any major school of thought, logical empiricism was fragmented, with many variations (Creath, 2014). Still, as a response to traditional idealism's preoccupation with abstract notions and doubts about any knowable reality outside the mind, logical empiricism (successor to logical positivism) limited the scope of scientific inquiry to that which could be observed. Empiricism denied that science could yield knowledge about anything not subject to observation (Hunt, 1991). Unobservable terms in a theory – a class that would include the unobservable conceptual variables in theoretical models – could be no more than labels for or representations of a pattern of empirical regularities (Dicken, 2016; Psillos, 1999). While these unobservable terms might be interpreted partly based on the context of their position in a theoretical model, there was also an "upward seepage of meaning from the observational terms to the theoretical concepts" (Feigl, 1970, p. 7). Even if some unobservable term did actually exist – just as germs existed before they could be observed – the scientific method could say nothing about it. As a result, "(…) claims about unobservables are meaningful for logical empiricists if and only if their unobservable terms are linked in an appropriate way to observable terms. The unobservable vocabulary is then treated as nothing more than a shorthand for the observation reports to which they are tied." (Chakravartty, 2007, p. 11). Alternatively, a theory that includes unobservable terms could be understood as a kind of metaphor, such that "… the 'cash value' of scientific theories is fully captured by what theories say about the observable world" (Psillos, 1999, p. xx).

Logical empiricism provided the inspiration for classical notions of construct validity. "Until the 1970s, the dominant epistemological home for understanding and testing for validity was the philosophy of logical empiricism." (Haig & Evers, 2016, p. xiii – see also Ch. 3). Classic works on validity embraced by the structural equation modeling community (Campbell & Fiske, 1959; Cronbach & Meehl, 1955) are avowedly empiricist. For example, Campbell (1960) strongly defended Cronbach and Meehl's (1955) "construct validity" against Bechtoldt's (1959) charge of *not* being empiricist. Cronbach (1989), looking back on the period, insisted that empiricism was the motivating philosophy. This approach can be read in Bollen's (1989) treatment of measurement validity. Validity indices, according to Bollen (1989), could be calculated as functions of factor model parameters, whose values were themselves based on observed data, without regard to any variables outside the given factor model. So it appears that empiricist thinking continued to influence the measurement validity practices associated with structural equation modeling.

The later twentieth century has seen the decline of empiricism (Creath, 2014; Fine, 1998; Keat, 1998; Passmore, 1967), and the rise of scientific realism (Borsboom, 2005; Chakravartty, 2015; Haig & Evers, 2016; Hunt, 1991). "Minimally speaking, realism involves a commitment to the ideas that there is a real world of which we are part, and that both the observable and unobservable features of that world can be known by the appropriate use of scientific methods" (Haig & Evers, 2016, p. 1). Scientific realism grants that important elements in even a mature scientific theory, like germs in the pre-microscope age, or subatomic particles even today, might be unobservable. These unobservable terms have their own existence, so they are not given meaning by observable terms. Unobserved conceptual variables, then, are genuine traits or aspects of individuals or groups, neither idealizations nor mere summarizations of empirical regularities. Realist researchers study a particular conceptual variable because they believe it to exist and to play a causal role, but this belief only affects the researchers' decision to include the conceptual variable within the scope of a research effort. In contrast with the contortions of the idealists, for the realist, beliefs do not affect the existence of the conceptual variable itself.

Of course, the scientific realist might be wrong about the unobserved conceptual variables at work – wrong about the character of the conceptual variables involved with a phenomenon, or about their very existence: "The possibility of error is explicitly acknowledged in the realist's commitment to fallibilism—the belief that our knowledge claims may be mistaken because human inquirers characteristically err" (Haig & Evers, 2016, p. 3). While being cognizant of that risk, it is within the scope of the scientific method for a researcher to structure investigations and pursue research questions as if a set of unobservable conceptual variables were real phenomena independent of observed variables. Statistical models, on the other hand, are a matter of data, and a researcher building a statistical model of the behavior of unobserved conceptual variables will use observed variables to build empirical representations of, or proxies for, the unobserved conceptual variables. For the realist user of structural equation modeling, the common factors in factor-based structural equation modeling (and the weighted composites in composite-based approaches) serve precisely this function. For such a researcher, discrepancies between conceptual variables and their empirical representations are an immediate and obvious confound, and questions about such discrepancies must be addressed with the best means available. One key distinction between empiricism and realism is that the latter asserts that the conceptual variable exists independently of the common factor that represents it in a factor model (Borsboom, 2005; Borsboom, Mellenbergh, & van Heerden, 2003; Markus & Borsboom, 2013).

MacCorquodale and Meehl (1948) introduced the notion of "hypothetical constructs," variables in theoretical models that have "excess meaning" beyond their direct empirical connections.[2] This notion seems very similar to the unobserved conceptual variables that a realist would study. In the examples they provide, however, MacCorquodale and Meehl (1948) and Cronbach and Meehl (1955) equate "hypothetical construct" and common factor. Cronbach and Meehl (1955) describe an evolving understanding of a "hypothetical construct" in terms of observed variables that do or do not load strongly together on a common factor. The performance of the factor model immediately defines the "hypothetical construct," in contrast to the realist view. When the empiricist is wrong about a common factor, the error is in misunderstanding the statistical communality, such that the empiricist makes wrong predictions about which other observed variables will or will not fit the factor model, if they are added to the current set of indicators. For the realist, beyond concerns about the statistical model, "wrong" means that the common factor does not behave in the same way as the unobserved conceptual variable. The empiricist's mistake is a matter of the behavior of a set of observed variables, while the realist's error references a reality beyond observed variables. Both the procedures and inferences described in examples provided by MacCorquodale and Meehl (1948) and by Cronbach and Meehl (1955), and the pains taken by Campbell (1960) and Cronbach (1989) to identify their work with an empiricist perspective suggest that they did not see an existence for their "hypothetical construct" that was independent of the common factor that represents it.

However, if the unobserved conceptual variable exists independently of the common factor, then factor indeterminacy acquires a new meaning. The common factor resides within the factor model, but the conceptual variable stands *outside* the model – unless, again, one takes an empiricist (or even operationist) perspective such that common factor and conceptual variable are tautologically equivalent (Borsboom, 2005, p. 13; Bridgman, 1927, p. 5; Chang, 2009).

Recall that factor indeterminacy creates uncertainty about the relation between a common factor within a model and *any* variable outside the model. If the conceptual variable represented by a common factor is itself outside the model, then factor indeterminacy induces uncertainty around the relationship between the common factor and the conceptual variable that the common factor represents. The coefficient $\kappa_{=} = 2(1 - \rho^2_{f.Y})$ then describes a range of uncertainty around the correlation between the factor and the conceptual variable – a range which can be quite wide when indicators are few and loadings are weak, as Tables 1–3 indicate. Of course, researchers would like the correlation itself to be strong. While that correlation itself is usually unknown, a factor model with high indeterminacy would make it impossible to tell precisely where the correlation actually lies. Given that conceptual space is full of a broad range of conceptual variables, correlated with each other to lesser or greater degrees, a wide uncertainty band would certainly raise doubt about just which conceptual variable is most closely associated with a given common factor. Even though factor indeterminacy has no effect on parameter estimates, such as path coefficients, within a factor model, the meaningfulness of those estimates is substantially diminished if the validity of the factors as representatives of particular conceptual variables is in question. Thus, factor indeterminacy should be taken as a crucial threat to validity in structural equation modeling – a threat that currently stands largely unacknowledged. Of course, if one rejects the realist position and thus rejects the independent existence of unobserved conceptual variables, then this aspect of the indeterminacy problem also does not exist.

Implications

The implications that follow from this perspective on factor indeterminacy, uncertainty, philosophy of

[2]We thank the associate editor for pointing us to this important reference.

science, and structural equation modeling are wide-ranging. Some of these are rather mechanical, suggesting variations in procedure but leaving current practice largely intact. Others call for a dramatic rethinking of the received view of structural equation modeling.

Acknowledging uncertainty in psychological measurement

If psychometricians, like metrologists, were to acknowledge uncertainty as a phenomenon distinct from systematic error and random error, then psychological measurement would need standard procedures for cataloging sources of uncertainty, the familiar sampling errors and factor indeterminacy being just a few of many. A focus on uncertainty sounds like a radical change, but this will not be the first call for a substantially new perspective: "The tidy theory of error laid down for psychology at the start of the century by Spearman and Brown has always seemed just a little too tidy to describe the perverse behavior of real data" (Cronbach, Gleser, Nanda, & Rajaratnam, 1972, p. v). *Every* academic field must be wary of complacency, of what Boker (2002, p. 420) called a "local minimum in theory space … ". Metrologists have responded to the challenge by agreeing on language and codifying procedures (e.g., Bucher, 2012; JCGM WG1, 2008; Tal, 2011), and by training a generation to follow those procedures, and their fields have been rewarded with reductions in uncertainty that have enabled a range of advances. Psychometrics could undertake the same.

Managing factor indeterminacy

If factor indeterminacy is a form of uncertainty, and researchers are convinced of the need to manage uncertainty, then they need to change the construction of factor models in order to limit indeterminacy to a tolerable level. Managing uncertainty surely means reducing it over time, though there is likely some degree of uncertainty that is judged acceptable for a particular purpose (Bucher, 2012; Goldsmith, 2010).

Factor indeterminacy is primarily a function of the number of observed variables per factor and the strength of factor loadings, but is secondarily affected by correlations between factors. For example, the combination of four indicators per factor with four correlated factors and with standardized factor loadings of 0.7 and factor correlations of 0.7 produces a $\kappa_{=}$ value of 0.329 (Table 3). Fewer indicators per factor and weaker loadings expand the range.

Among professional testing firms, such as those that produce standardized admissions or certification tests, the common practice is to use many indicators. Even if some loadings are weak, and individual items even load on multiple factors, factor indeterminacy is probably low. Among publications in academic disciplines, however (and without conducting an actual survey of publications), most applications of structural equation modeling report fewer indicators. There are good reasons for this. Besides the additional direct expense of developing and administering longer instruments, there is the increased risk of respondent fatigue and incomplete data – professional testing firms employ adaptive testing procedures, with fewer items, for similar reasons. And researchers may well start a project with more indicators in mind but then trim the number actually used in a model because some of them perform poorly or appear to be misunderstood by respondents. Another reason – potentially the most fundamental – is researchers' urge to achieve adequate model fit, as assessed by criteria like the χ^2 statistic or related alternative fit indices, because publication success is often tied to model fit. While having more observed variables per common factor reduces factor indeterminacy, and hence uncertainty (Tables 1–3), it also contributes to worse fit in multiple ways. More observed variables per factor means more degrees of freedom, which means more power (MacCallum, Browne, & Sugawara, 1996). More indicators also means more opportunities for highly constrained factor models to fail:

> Psychometric indicators are seldom perfectly pure construct indicators. (…) Although 'pure' indicators of a single construct may exist, we surmise that such indicators remain at best a convenient fiction and that, in practice, most indicators will present both some level of random noise and some level of construct-relevant association with other constructs (…). (Asparouhov, Muthén, & Morin, 2015, p. 1563)

Therefore, researchers may face a tradeoff between model fit and uncertainty. Including many indicators for every common factor, in pursuit of reduced uncertainty, invites the discovery of commonalities across items that are not accounted for by the model. Perhaps the need to consider such a tradeoff will spark useful discussions about the actual importance of fit vs. uncertainty in structural equation modeling.

Relatedly, researchers also care about scoring above minimum "cut-off" values on reliability criteria like composite reliability. Succeeding against such criteria argues for stronger loadings, just as managing factor indeterminacy does. But achieving sufficiently small values for $\kappa_{=}$ is likely to mean researchers must use *both* more indicators and indicators with stronger loadings. Researchers may also need to think differently about those poorly performing indicators. While researchers may be all too willing to cull items with weak loadings to improve reliability indices (as long as the minimum number of items remain), this culling may increase factor indeterminacy and uncertainty.

Of course, researchers who create "factor scores" need to take account of factor indeterminacy. If indeterminacy is high, researchers need to be exceptionally cautious in the interpretation of any results employing the scores. Researchers can reduce indeterminacy by calculating factor scores as part of a model containing multiple common factors, if the factors are correlated. Correlations among the factors marginally reduce the indeterminacy of each factor. Perhaps more substantially, factor indeterminacy does *not* contribute uncertainty to correlations among factors *within* the same factor model. By contrast, researchers who construct factor scores using separate factor models for each common factor *induce* uncertainty regarding relations between each and every factor, never mind uncertainty regarding relations between the various factors and external variables. Researchers who need factor scores should use all available information in their calculation to minimize the influence of the factors' arbitrary components.

Evaluating construct validity

As noted previously, in structural equation modeling, the standard approach for assessing construct validity involves computing indices that are functions of factor model parameter estimates. Researchers use this "validity" of the common factors as an endorsement for statements about conceptual variables in theoretical models. But if "validity" is defined only in terms of factor model parameters, then its scope should be limited to the factors themselves, only justifying statements about relations among the common factors in the statistical model. This "validity" would say nothing about relations among the conceptual variables that the common factors represent.

From an empiricist perspective, ignoring the conceptual variables makes sense, because either they have no independent existence – theoretical models that mention them are all a sort of metaphor, not to be taken literally – or else they lie beyond the reach of the scientific method. From a realist perspective, the theoretical model is a literal statement of what the scientist believes, based on available evidence. But even then, one might logically focus on the common factors and ignore the conceptual variables if common factor and conceptual variable were identical or equivalent.

One might choose to act as if common factor and conceptual variable are equivalent simply because the foreseeable problems from recognizing a distinction are just too great to overcome. Or one might be lured into believing in such an equivalence. Bollen's (1989) classic text introduces structural equation modeling by first discounting alternative approaches that effectively link conceptual variables directly to observed variables. Other approaches such as path analysis, Bollen (1989) notes, erroneously imply that *observed variable* and conceptual variable are identical, when it seems more plausible to believe that any given observed variable is tainted with error. In Bollen's (1989) treatment, acknowledging the presence of error in observed variables leads naturally to the adoption of structural equation modeling. Recommending structural equation modeling as a better alternative certainly suggests that the common factors in structural equation modeling are – at least in principle – not themselves tainted, but rather are equivalent to the conceptual variables they represent. The language in published structural equation modeling applications breezily slips from discussing conceptual variables to discussing common factors, and back again, as if the two were identical.

It must be noted that users of composite-based methods (e.g., Hair, Hult, Ringle, Sarstedt, & Thiele, 2017; Rigdon, 2013) do precisely the same thing, even knowing that their composites, being composed of error-tainted observed variables, cannot possibly claim such an equivalence. An empiricist stance is at least as common – and arguably is even more so – among users of composite-based methods, and the consequences for researcher behavior are largely the same.

Moreover, there has always been resistance to this tendency within the factor model literature. Cliff (1983, p. 120–123) and de Leeuw and Nishisato (1985, p. 373) warned against the "nominalist" or "naming" fallacy, respectively – the false notion that assigning a label to a common factor somehow made the factor identical with a particular trait or attribute. Markus and Borsboom (2013, p. 11) echo prior warnings about "reification" of common factors. They point to counter-arguments, such as what may be called the "rotation argument." Factor model solutions are not in general rotationally unique – a given solution can be rotated to new solutions, producing new sets of factors.[3] The researcher reduces rotational indeterminacy by imposing constraints based on prior knowledge – prior knowledge about the observed variables in the model or prior knowledge about the conceptual variables which the common factors are expected to mimic.

Yet, the solution to rotational indeterminacy – using prior information – suggests an approach to assessing construct validity that can accommodate an awareness of the discrepancy between common factor and conceptual variable and that does not rely entirely on information internal to the factor model itself. In his Messenger lectures at Cornell University, the physicist Richard Feynman spoke about the rigors of scientific research: "What we need is imagination, but imagination in a terrible straitjacket. We have to find a new view of the world that has to agree with everything that is known." (Feynman, 1965/1994, p. 165). The validity of common factors as representations of conceptual variables can be evaluated if researchers use whatever is already known about the conceptual variables in question.

For the realist, if there is a genuine conceptual variable with its own existence, then that conceptual variable has its own behavior. That behavior may be conditional on other circumstances, but still, it has a certain behavior. Researchers who use a common factor to represent that conceptual variable should require the factor to behave like the conceptual variable. Structural equation modeling researchers have already done this (to a limited degree) when specifying growth models and interaction models, where constraints on parameter estimates, for example, force a common factor to behave like a linear slope, or like a product of two other common factors. If a conceptual variable is only as real as that, then it is not asking too much that researchers constrain the behavior of their common factor to similarly conform to the conceptual variable's behavior.

The requirements for enabling this proposal are actually quite modest. First, researchers need to catalog what is believed to be known about each conceptual variable. Then researchers need to build their factor models incrementally, incorporating elements that are known along with elements which remain unknown – and constraining the "known" parts to conform to what is known. Evaluating measures and evaluating theoretical models have always been two interrelated tasks (Schwab, 1999). If the researcher can assume that the theoretical model is correct, then measurement can be evaluated by determining the extent to which measures yield results consistent with theory. If measurement can be assumed valid, then theoretical models can be evaluated by comparing theoretical predictions with empirical results. When the status of both theory and measurement are unknown, neither can be evaluated with confidence. So there must be "knowns." The change proposed here is that this prior knowledge actually be used to force the statistical behavior of the common factors to mirror the theoretical behavior of the conceptual variables that the factors represent, thus enabling an approach to validity assessment that acknowledges a distinction between common factor and conceptual variable. Indeed, returning to the discussion of the formula for $\kappa_{=}$, "knowns" might eventually support a researcher claiming that $\rho^2_{\omega.Y} > \rho^2_{f.Y}$, in a given instance – because a function of observed variables so accurately reproduces known-correct relations with a set of conceptual variables – thus further shrinking estimated uncertainty. Arguing that $\rho^2_{\omega.Y}$ is higher than $\rho^2_{f.Y}$ suggests that a composite can capture the conceptual variable more accurately than the common factor. This would question the use of the common factor model in the first place.

Method comparisons

Recognizing factor indeterminacy as uncertainty also suggests reexamination of the debate regarding the relative merits of common factor-based methods vs. composite-based methods in structural equation modeling. A full consideration of this debate is beyond this paper's scope, but recognizing factor indeterminacy as uncertainty, and recognizing uncertainty as something to be managed and minimized, changes the nature of the conversation. It has been standard to argue that the factor-based approach "accounts for measurement error." By this is meant that the factor model includes "error terms" (unique factors) for the observed variables (the observed variables being modeled as dependent on the common factors). Composite-based methods, such as partial least squares path modeling (e.g., Hair, Sarstedt, Ringle, & Gudergan, 2018; Rigdon, 2013) and generalized structured component analysis (e.g., Hwang & Takane, 2015) have lacked these parameters (but see Hwang, Takane, & Jung, 2017, for an exception), since the composites in these methods are formed out of the observed variables.

[3]Note that rotational indeterminacy and factor indeterminacy are not the same thing. Factor indeterminacy will remain even if all model parameters are fixed at particular values and the entire solution is thus fixed, as long as the rank problem remains.

What is called "measurement error" in factor models is directly tied to factor indeterminacy. The larger the observed variables' residual variances, the smaller the standardized loadings, and the greater the factor indeterminacy, all other things equal. Recognizing factor indeterminacy as uncertainty reveals a new perspective. These residual variances are not harmlessly neutralized; rather, they are converted into uncertainty.

Researchers would derive substantial benefit from a full accounting, and a fresh debate, relating to the compromises involved in using either common factors or weighted composites as stand-ins or proxies for conceptual variables. There are longstanding claims of advantage for one general approach or the other, which might profitably be reexamined. It has been almost 30 years since a special section of *Multivariate Behavioral Research* hosted Velicer and Jackson's (1990) focus paper and ensuing commentaries by renowned authors on the relative merits of factor-based and composite-based methods – three decades full of innovations and insights (Widaman, 2018). Moreover, the metrology perspective may provide a true unifying framework for thinking about measurement in a way that is not itself inherently bound to one statistical method or another.

While this paper has made the point that factor indeterminacy is a form of metrological uncertainty, this is not meant to imply that factor models are subject to uncertainty while composite-based methods are not. The *Guide to the Expression of Uncertainty* (Section 3.3.2) identifies many potential contributors to overall uncertainty besides random sampling error, including incomplete definition of the measurand, non-representative sampling, inadequate knowledge of the effects of environmental conditions on the measurement, and many more.

This presentation has sought to make one central point. To that end, this presentation assumed that population-level data were available, that the estimated factor model was correct in the population, and that the researcher was correct about the correspondence between common factor and unobserved variable. As the associate editor pointed out during the review process, in practice, these additional sources of uncertainty should also be included in the assessment of overall uncertainty. From the beginning of structural equation modeling as inferential method, researchers have stressed that constrained models do not fit exactly (Jöreskog, 1969, p. 200). The problem of model selection uncertainty in factor-based structural equation modeling has also begun to be addressed (Preacher & Merkle, 2012). The $\kappa_{=}$ statistic presented here must be taken as a best case for the degree of uncertainty regarding the correlation between a common factor and any excluded variable, including the unobserved conceptual variable for which the common factor proxies.

Accounting for the uncertainty associated with a research conclusion means accounting of *all* material contributions to that uncertainty, regardless of whether using composite- or common factor-based methods. This paper should not be taken as suggesting that composite-based methods have an overall advantage in terms of uncertainty, just because they currently have no ready way of calculating uncertainty of the relations between component scores and conceptual variables. Rather, while the factor model indeed includes a procedure for "accounting for measurement error" – something long cited as an advantage for factor methods – that very procedure induces its own contribution to uncertainty, in the form of factor indeterminacy.

Discussion

Despite decades of interest, structural equation modeling has largely ignored factor indeterminacy. It has been easy to do this, partly because there was no place to put the indeterminacy phenomenon in psychometrics' narrow typology of error. Moreover, reliance on an empiricist philosophy of science would mean that researchers need not concern themselves with unobserved conceptual variables that are distinct and independent from the common factors (or weighted composites) in statistical models. Borsboom (2005), Borsboom et al. (2003) and Markus and Borsboom (2013) have outlined scientific realist perspectives that opened the door to considering conceptual variables in their own right, but without a problem to solve, it has been easy to conclude that any differences in perspective are of little consequence – as Cronbach (1989) observed.

If, however, factor indeterminacy is a manifestation of what metrologists call uncertainty, and if reducing uncertainty in measurement helps to move scientific research forward, then it may be time to take factor indeterminacy seriously, and also time to consider expanding the psychometric framework to include uncertainty in measurement.

It is not as if behavioral research has been a pointless exercise. The field made tremendous progress in the early 1900s in putting itself on firmer quantitative footing. And the rapid spread of structural equation modeling in the later 1900s (e.g., Babin, Hair, & Boles, 2008; Bentler, 1986; Hair, Sarstedt, Ringle, & Mena, 2012) enabled new kinds of research and new insights. But the rapid progress in those periods may have obscured lurking problems, and maybe now it is time to address those problems. Maybe there is finally a place for factor indeterminacy in the frameworks used to evaluate research results, and a place for metrological uncertainty in the design of behavioral research. For example, Campbell and Fiske's (1959; also see Widaman, 1985) procedures for assessing discriminant validity involved the use of fundamentally different methods for tapping the same domain within the same study; however, applications of structural equation modeling are often monomethod (Voorhees, Brady, Calantone, & Ramirez, 2016). In the dominant psychometric framework, anything which is a constant, like the use of a single research method, does not contribute to error variance, though it certainly can induce bias. But that bias will be hard to detect, and against the costs and risks of actually employing multiple distinct research approaches, it is easy for researchers to "play it safe" and use a single method. In metrology, every aspect of a research design could potentially make a material contribution to uncertainty, even when that facet is a constant throughout the study. So metrology's focus on uncertainty might help to prevent researchers from "gaming the system" and shirking their responsibility to rigorously test research hypotheses.

Incorporating metrological concepts within psychometrics likely won't be easy – the metrology community disdains behavioral research, being hardly willing to go as far as "soft metrology" (Pointer, 2003) – but the goal was never to do things the easy way. Rather, the goal of behavioral research has always been to expand the boundaries of knowledge, confronting the critical challenges along the way.

Article Information

Conflict of interest disclosures: Each author signed a form for disclosure of potential conflicts of interest. Jan-Michael Becker is a co-developer and co-founder of SmartPLS (www.smartpls.com). SmartPLS is a statistical software for modeling and analyzing partial least squares path models. The other authors did not report any financial or other conflicts of interest in relation to the work described.

Ethical principles: The authors affirm having followed professional ethical guidelines in preparing this work. These guidelines include obtaining informed consent from human participants, maintaining ethical treatment and respect for the rights of human or animal participants, and ensuring the privacy of participants and their data, such as ensuring that individual participants cannot be identified in reported results or from publicly available original or archival data.

Funding: This work was not supported.

Role of the funders/sponsors: None of the funders or sponsors of this research had any role in the design and conduct of the study; collection, management, analysis, and interpretation of data; preparation, review, or approval of the manuscript; or decision to submit the manuscript for publication.

Acknowledgments: The authors thank Donald Schopflocher, Keith Markus, three exceptional reviewers and the editors for their comments on prior versions of this manuscript. The ideas and opinions expressed herein are those of the authors alone, and endorsement by the authors' institutions is not intended and should not be inferred.

References

Asparouhov, T., Muthén, B., & Morin, A. J. S. (2015). Bayesian structural equation modeling with cross-loadings and residual covariances: Comments on Stromeyer. *Journal of Management*, *41*(6), 1561–1577. doi: 10.1177/0149206315591075.

Babin, B. J., Hair, J. F., & Boles, J. S. (2008). Publishing research in marketing journals using structural equation modeling. *Journal of Marketing Theory & Practice*, *16*(4), 279–285. doi: 10.2753/MTP1069-6679160401.

Bechtoldt, H. P. (1959). Construct validity: A critique. *The American Psychologist*, *14*(10), 619–629. doi: 10.1037/h0040359.

Bell, S. (1999). *Good practice guide #12: A beginner's guide to uncertainty of measurement*. Teddington, UK: National Physical Laboratory.

Bentler, P. M. (1986). Structural modeling and Psychometrika: An historical perspective on growth and achievements. *Psychometrika*, *51*(1), 35–51. doi: 10.1007/BF02293997.

Bollen, K. A. (1989). *Structural equations with latent variables*. New York, NJ: John Wiley & Sons.

Boker, S. M. (2002). Consequences of continuity: The hunt for intrinsic properties within parameters of dynamics in psychological processes. *Multivariate Behavioral Research*, *37*(3), 405–422. doi: 10.1207/S15327906MBR3703_5.

Borsboom, D. (2005). *Measuring the mind: Conceptual issues in contemporary psychometrics*. Cambridge, UK: Cambridge University.

Borsboom, D., Mellenbergh, G. J., & van Heerden, J. (2003). The theoretical status of latent variables. *Psychological Review*, *110*(2), 203–219. doi: 10.1037/0033-295X.110.2.203.

Bridgman, P. W. (1927). *The logic of modern physics*. New York: MacMillan.

Bucher, J. L. (2012). *The metrology handbook*. Milwaukee, WS: Quality Press.

Campbell, D. T. (1960). Recommendations for APA test standards regarding construct, trait, or discriminant validity. *The American Psychologist*, *15*(8), 546–553. doi: 10.1037/h0048255.

Campbell, D. T., & Fiske, D. W. (1959). Convergent and discriminant validation by the multitrait-multimethod matrix. *Psychological Bulletin*, *56*(2), 81–105. doi: 10.1037/h0046016.

Chakravartty, A. (2007). *A metaphysics for scientific realism: Knowing the unobservable*. Cambridge, MA: Cambridge University.

Chakravartty, A. (2015). Scientific realism. In E. N. Zalta (Ed.), *Stanford encyclopedia of philosophy*. Retrieved from http://plato.stanford.edu/archives/fall2015/entries/scientific-realism/. Accessed 02/10/2017.

Chang, H. (2009). Operationalism. In E. N. Zalta (Ed.), *Stanford encyclopedia of philosophy*. Retrieved from http://plato.stanford.edu/archives/fall2009/entries/operationalism/. Accessed 02/10/2017.

Chou, C. W., Hume, D. B., Rosenband, T., & Wineland, D. J. (2010). Optical clocks and relativity. *Science*, *329*(5999), 1630–1632. doi: 10.1126/science.1192720.

Cliff, N. (1983). Some cautions concerning the application of causal modeling methods. *Multivariate Behavioral Research*, *18*(1), 115–126. doi: 10.1207/s15327906mbr1801_7.

Cohen, J., Cohen, P., West, S. G., & Aiken, L. S. (2003). *Applied multiple regression/correlation analysis for the behavioral sciences* (3rd ed.). Mahwah, NJ: LEA.

Creath, R. (2014). Logical empiricism. In E. N. Zalta (Ed.), *Stanford encyclopedia of philosophy*. Retrieved from http://plato.stanford.edu/archives/spr2014/entries/logical-empiricism/. Accessed 2/10/2017.

Cronbach, L. J. (1989). Construct validation after thirty years. In R. L. Linn (Ed.), *Intelligence: Measurement, theory and public policy* (pp. 147–171). Urbana, IL: University of Illinois.

Cronbach, L. J., Gleser, G. C., Nanda, H., & Rajaratnam, N. (1972). *The dependability of behavioral measurements: Theory of generalizability for scores and profiles*. New York, NY: Wiley.

Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological Bulletin*, *52*(4), 281–302. doi: 10.1037/h0040957.

de Leeuw, J., & Nishisato, S. (1985). Reviews. *Psychometrika*, *50*(3), 371–375. doi: 10.1207/S15328007SEM0701_08.

Dicken, P. (2016). *A critical introduction to scientific realism*. London: Bloomsbury.

Feigl, H. (1970). The 'orthodox' view of theories: Remarks in defense as well as critique. In M. Radner, & S. Winokur (Eds.), *Minnesota studies in the philosophy of science* (Vol. 4, pp. 3–16). Minneapolis, MN: University of Minnesota Press.

Feynman, R. (1965/1994). *The character of physical law*. New York, NY: The Modern Library.

Fine, A. (1998). Scientific realism and antirealism, 1998. doi: 10.4324/9780415249126-Q094-1. Routledge Encyclopedia of Philosophy, Taylor and Francis. Retrieved from https://www.rep.routledge.com/articles/thematic/scientific-realism-and-antirealism/v-1.

Gibney, E. (2015). Atomic clocks face off. *Nature*, *522*(7554), 16–17. doi: 10.1038/522016a.

Goldsmith, M. (2010). *Good practice guide #118: A beginner's guide to measurement*. Teddington, UK: National Physical Laboratory.

Grice, J. W. (2001). Computing and evaluating factor scores. *Psychological Methods*, *6*(4), 430–450. doi: 10.1037/1082-989X.6.4.430.

Guttman, L. (1955). The determinacy of factor score matrices with implications for five other basic problems of common-factor theory. *British Journal of Statistical Psychology*, *8*(2), 65–81. doi: 10.1111/j.2044-8317.1955.tb00321.x.

Haig, B. D., & Evers, C. W. (2016). *Realist inquiry in social science*. London: Sage.

Hair, J. F., Hult, G. T. M., Ringle, C. M., Sarstedt, M., & Thiele, K. O. (2017). Mirror, mirror on the wall: a comparative evaluation of composite-based structural equation modeling methods. *Journal of the Academy of Marketing Science*, *45*(5), 616–632. doi: 10.1007/s11747-017-0517-x.

Hair, J. F., Sarstedt, M., Ringle, C. M., & Gudergan, S. P. (2018). *Advanced issues in partial least squares structural equation modeling*. Thousand Oaks, CA: Sage.

Hair, J. F., Sarstedt, M., Ringle, C. M., & Mena, J. A. (2012). An assessment of the use of partial least squares structural equation modeling in marketing research. *Journal of the Academy of Marketing Science*, *40*(3), 414–433. doi: 10.1007/s11747-011-0261-6.

Higgins, K., Miner, D., Smith, C. N., & Sullivan, D. B. (2004). *A walk through time* (version 1.2.1). [Online] Retrieved from http://physics.nist.gov/time [2010, July 12]. Gaithersburg, MD: National Institute of Standards and Technology.

Hoyle, R. H. (Ed.). (2014). *Handbook of structural equation modeling*. New York: Guilford Press.

Hunt, S. D. (1991). *Modern marketing theory: Critical issues in the philosophy of marketing science*. Cincinnati, OH: South-Western.

Hwang, H., & Takane, Y. (2015). *Generalized structured component analysis: A component-based approach to structural equation modeling*. Boca Ratron, FL: CRC Press.

Hwang, H., Takane, Y., & Jung, K. (2017). Generalized structured component analysis with uniqueness terms for accommodating measurement error. *Frontiers in Psychology*, *8*, 21–37. doi: 10.3389/fpsyg.2017.02137.

Joint Committee for Guides in Metrology/Working Group on the Expression of Uncertainty in Measurement (JCGM/WG1). (2008). *Evaluation of measurement data—Guide to the expression of uncertainty in measurement*. Retrieved from http://www.iso.org/sites/JCGM/GUM-introduction.htm. Accessed 02/10/2017.

Jöreskog, K. G. (1969). A general approach to confirmatory maximum likelihood factor analysis. *Psychometrika*, *34*(2), 183–202. doi:10.1007/BF02289343

Keat, R. (1998). Scientific realism and social science, 1998, doi: 10.4324/9780415249126-R025-1. Routledge Encyclopedia of Philosophy, Taylor and Francis. Retrieved from https://www.rep.routledge.com/articles/thematic/scientific-realism-and-social-science/v-1.

MacCallum, R. C., Browne, M. W., & Sugawara, H. M. (1996). Power analysis and determination of sample size for covariance structure modeling. *Psychological Methods*, *1*(2), 130–149. doi: 10.1037/1082-989X.1.2.130.

MacCorquodale, K., & Meehl, P. E. (1948). On a distinction between hypothetical constructs and intervening variables. *Psychological Review*, *55*(2), 95–107. doi: 10.1037/h0056029.

Maraun, M. D. (1996). Metaphor taken as math: Indeterminacy in the factor analysis model. *Multivariate Behavioral Research*, *31*(4), 517–538. doi: 10.1207/s15327906mbr3104_6.

Maraun, M. D., & Gabriel, S. M. (2013). Illegitimate concept-equating in the partial fusion of construct validation theory and latent variable modeling. *New Ideas in Psychology*, *31*(1), 32–42. doi: 10.1016/j.newideapsych.2011.02.006.

Markus, K. A., & Borsboom, D. (2013). *Frontiers in test validity theory. Measurement, causation, and meaning*. New York, NY: Routledge.

Mayo, D. G. (1981). In defense of the Neyman-Pearson theory of confidence intervals. *Philosophy of Science*, *48*(2), 269–280. doi: 10.1086/288996.

McDonald, R. P. (1999). *Test theory: A unified treatment*. Mahwah, NJ: LEA.

Michell, J. (1999). *Measurement in psychology: A critical history of a methodological concept*. Cambridge: Cambridge University Press.

Michell, J. (2013). Constructs, inferences and mental measurement. *New Ideas in Psychology*, *31*(1), 13–21. doi: 10.1016/j.newideapsych.2011.02.004

Mulaik, S. A. (2010). *Foundations of factor analysis* (2nd ed.). Boca Raton, FL: Chapman & Hall.

Mulaik, S. A., & McDonald, R. P. (1978). The effect of additional variables on factor indeterminacy in models with a single common factor. *Psychometrika*, *43*(2), 177–192. doi: 10.1007/BF02293861.

Passmore, J. (1967). Logical positivism. In P. Edwards (Ed.), *The encyclopedia of philosophy* (pp. 52–57). New York, NY: Macmillan.

Pointer, M. R. (2003). *New directions—soft metrology requirements for support from mathematics, statistics and software*. Teddigton: National Physical Laboratory. Retrieved from http://publications.npl.co.uk/npl_web/pdf/cmsc20.pdf. Accessed 29/12/2017.

Preacher, K. J., & Merkle, E. C. (2012). The problem of model selection uncertainty in structural equation modeling. *Psychological Methods*, *17*(1), 1–14. doi: 10.1037/a0026804.

Psillos, S. (1999). *Scientific realism: How science tracks truth*. Abingdon: Routledge.

Rigdon, E. E. (1994). Demonstrating the effects of unmodeled random measurement error. *Structural Equation Modeling*, *1*(4), 375–380. doi: 10.1080/10705519409539986.

Rigdon, E. E. (2013). Partial least squares path modeling. In G. Hancock & R. Mueller (Eds.), *Structural equation modeling: A second course* (2nd ed., pp. 81–116). Charlotte, NC: Information Age.

Rosenband, T., Hume, D. B., Schmidt, P. O., Chou, C. W., Brusch, A., Lorini, L., … Bergquist, J. C. (2008). Frequency ratio of Al^+ and Hg^+ single-ion optical clocks: metrology at the 17th decimal place. *Science*, *319*(5871), 1808–1812. doi: 10.1126/science.1154622.

Schönemann, P. H., & Steiger, J. H. (1976). Regression component analysis. *British Journal of Mathematical and Statistical Psychology*, *29*(2), 175–189. doi: 10.1111/j.2044-8317.1976.tb00713.x.

Schönemann, P. H., & Steiger, J. H. (1978). On the validity of indeterminate factor scores. *Bulletin of the Psychonomic Society*, *12*(4), 287–290. doi: 10.3758/BF03329685.

Schönemann, P. H., & Wang, M.-M. (1972). Some new results on factor indeterminacy. *Psychometrika*, *37*(1), 61–91. doi: 10.1007/BF02291413.

Schwab, D. P. (1999). *Research methods for organizational studies*. Mahwah, NJ: Lawrence Erlbaum Associates.

Slaev, V. A., Chunovkina, A. G., & Mironovsky, L. A. (2013). *Metrology and theory of measurement*. Berlin: Walter de Gruyter.

Spearman, C. (1904). "General intelligence," objectively determined and measured. *The American Journal of Psychology*, *15*(2), 201–292. doi: 10.2307/1412107.

Steiger, J. H. (1979a). The relationship between external variables and common factors. *Psychometrika*, *44*(1), 93–97. doi: 10.1007/BF02293788.

Steiger, J. H. (1979b). Factor indeterminacy in the 1930's and the 1970's some interesting parallels. *Psychometrika*, *44*(2), 157–167. doi: 10.1007/BF02293967.

Tal, E. (2011). How accurate is the standard second? *Philosophy of Science*, *78*(5), 1082–1096. doi: 10.1086/662268.

Velicer, W. F., & Jackson, D. N. (1990). Component analysis versus common factor analysis: Some issues in selecting an appropriate procedure. *Multivariate Behavioral Research*, *25*(1), 1–28. doi: 10.1207/s15327906mbr2501_1.

Voorhees, C. M., Brady, M. K., Calantone, R., & Ramirez, E. (2016). Discriminant validity testing in marketing: An analysis, causes for concern and proposed remedies. *Journal of the Academy of Marketing Science*, *44*(1), 119–134. doi: 10.1007/s11747-015-0455-4.

Widaman, K. F. (1985). Hierarchically nested covariance structure models for multitrait-multimethod data. *Applied Psychological Measurement*, *9*(1), 1–26. doi: 10.1177/014662168500900101.

Widaman, K. F. (2018). On common factor and principal component representations of data: Implications for theory and for confirmatory replications. *Structural Equation Modeling: A Multidisciplinary Journal*, *25*(6), 829–847. doi: 10.1080/10705511.2018.1478730

Wilson, E. B. (1928). The abilities of man: Their nature and measurement. *Science*, *67*(1731), 244–248. doi: 10.1126/science.68.1750.38-a.